Week 2 - Learning Map

Goal:

Implement a semantic retrieval layer for SupportDesk RAG: embed tickets and queries, store vectors with metadata, chunk and index offline, then retrieve Top-K evidence online.

Validate relevance with test queries, tune chunking, and ensure outputs are grounded, repeatable, and production-ready.

Overview...

Week 2 focuses on building the retrieval layer of a SupportDesk RAG assistant so it can quickly surface the best historical support tickets for a new issue.

You will learn why keyword search fails on unstructured text and synonyms, then implement semantic retrieval using text embeddings and a vector database.

Next, you will practice chunking long tickets into retrievable units—fixed-size, semantic, recursive, structure-based, and LLM-assisted approaches—so retrieval stays precise without losing context.

You will run an offline indexing pipeline that stores each chunk's embedding plus metadata, avoiding costly full scans at query time.

Finally, you will execute the core retrieval loop: convert a question to a searchable representation, fetch Top-K candidates, optionally rerank, and pass the best context to the model for grounded answers.

You'll see why context limits require selective retrieval and how multiple indexes support routing.

By the end, you'll have working demos, tuning levers, and a clear path into Week 3's RAG pipeline, evaluation, and agentic RAG.

Note:

Post-class resources are optional, but the pre-class setup must be completed before the live class.

HOW TO START?

Roadmap

Pre-class Setup

Setup + Problem Framing
- API key + env setup
- Clone repo
- Understand the goal: retrieve relevant past support tickets quickly & reliably

Live Class

Embeddings + Vector Store

- Text → vectors (embeddings)
- Store vectors + metadata in vector DB
- Semantic search: query → Top-K matches

Chunking + Indexing (Offline pipeline)

- Chunk long tickets (fixed/semantic/...)
- Create index: chunks + embeddings + metadata
- Prep for scale + consistent retrieval

Retrieval Loop + Validation

- Question → embed → retrieve Top-K
- (Optional) rerank → send best context
- Test queries; tune chunking/indexing
- Preview: Week 3 = RAG pipeline + evaluation

Learners are encouraged to follow along with the instructor during the live session and execute the code on their local environment.

Post-class resources (Optional deep dives)

- Vector Store
- Embedding + Vector Database + Pinecone_VectorDB Notebook
- Access control for Vector Databases
- Chunking

Assignment

- Exercises - Embeddings, Chunking & Indexing

Please use the Wednesday AMA session to clarify any doubts you have on the topics. The assignment solutions will be discussed during the Saturday Assignment Review session.

All the resources have been added to your Uplevel account. If you have any questions, please reach out to the Operations team or raise a support ticket.